

From detector signals to $\Delta(x,Q^2)$: the reconstruction chain of the inclusive and coherent $\cos 2\phi$ measurements

2026-08-24 · revised 2026-08-26 · companion to the $\cos 2\phi$ projection report (money plots 5–7) and to plans/07 WP3 · computed with `polligen.reco`, `polligen.hfs` and `polligen.recopseudo` (`evgen/scripts/` : `reco_chain_figures.py`, `hfs_resolution.py`, `money_cos2phi_reco.py`, `money_cos2phi_coherent_reco.py`) on the PYTHIA 8 hadronic final state of `tools/pythia8` · revision history in Appendix A

The projection report presents the double-helicity-flip structure function $\Delta(x,Q^2)$ as ϕ' pseudo-data and as extracted $x\Delta$ points. This note is the chain behind them: what the detector records event by event in the inclusive and in the coherent intact-⁶Li measurement, how the azimuth carrying the observable is built from the reconstructed four-vectors and the known spin direction, which kinematic reconstruction assigns (x, Q^2) , which estimator turns spin-sorted ϕ' histograms into an amplitude, how the amplitude becomes Δ , and how well the chain closes at the reconstructed level (§7).

Three findings that change the analysis. (1) Three of the four inclusive sweet-spot bins of money plot 5 sit at $y = 0.010\text{--}0.025$, where the electron method gives $\delta y/y = 50\text{--}120\%$ even with a 1.2% calorimeter; x must come from the hadronic final state. On the PYTHIA 8 final state with 50 MeV of calorimeter noise the Σ method gives $\delta y/y = 0.32 / 0.22 / 0.28 / 0.11$ at the four spots; the reconstructed bins keep 40–73% purity and their amplitude sits 4–21% below the true-bin value, a response worth fitting rather than assuming (§7). (2) The single-fill $\cos 2\phi'$ fit measures the detector's own $\cos 2\phi'$ acceptance harmonic divided by P_{zz} : a 3% harmonic aligned with the vertical spin axis — the symmetry axis of the crossing angle and of the Roman-Pot cutout alike — fakes an amplitude four times the signal. The spin-state-sorted ratio of $m = \pm 1$ -rich and $m = 0$ -rich fills cancels any common acceptance bin by bin, turns relative-luminosity errors into a constant, and is $1.5\times$ *more* precise. (3) In the coherent channel the anchored deformation term modulates the *recoil* azimuth $\phi_t - \phi_S$, not the electron azimuth of the projection; and the Roman-Pot aperture, measured with an intact ⁶Li through the ePIC geometry, opens *horizontally*, fakes $\langle \cos 2\beta \rangle = -0.77$ against a physics $a_2 \approx 0.036$, and leaves 1.4% of the coherent recoils at 20.5 GeV/u, 5×10^{-5} at 50 and no accepted recoil at all in the binned window at 50 or 137.5: a low-energy programme (§4, §7).

1 · What is measured

Both measurements share the trigger object — a scattered electron in the central detector — and the spin bookkeeping of the ion beam; the coherent one adds the intact ⁶Li in the far-forward spectrometers and an exclusivity selection. The observable is a *Fourier coefficient of a ratio* of spin-sorted azimuthal distributions; Δ follows from it through kinematic factors and the unpolarized structure functions of the same data.

RECORDED PER EVENT	INCLUSIVE E ⁶ LI → E' X	COHERENT E ⁶ LI → E' X ⁶ LI(G.S.)
Beam and spin	Bunch-crossing ID → spin state ($m = \pm 1$ -rich / $m = 0$ -rich pattern); P_{zz} per fill from the polarimeter (scale $\approx 3\%$); alignment axis $\hat{n}$, vertical at IP6, with its measured tilt; luminosity per spin state.	
Central detector	e' candidate: track (p, θ, ϕ at the vertex) matched to an EMCal cluster, with E/p and shower shape for electron ID; the hadronic final state (all other tracks and clusters) for $\Sigma = \Sigma_h(E_h - p_{z,h})$; vertex.	The same, plus the diffractive system X (its mass M_X and the rapidity gap) — the only handle on x_p , since the recoil momentum is unresolved.
Far forward	Not required (breakup fragments of ⁶ Li are mostly beam-blind at IP6; plans/00).	Roman-Pot track at beam rigidity ($ R - 1 < 0.05$) with transverse displacement above the 10σ envelope (§4 step 1); vetoes in OMD ($p, {}^3\text{He}$), ZDC (n), B0 EMCal (de-excitation γ); Z-identification if available (open question #19).
Observable	$\cos 2\phi'$ coefficient of the ratio of spin-sorted $N_f(\phi')$ in (x, Q^2) bins, $\phi' = \phi_e - \phi_S$	$\cos 2\alpha, \cos 2\beta$ and $\cos(\alpha+\beta)$ coefficients of the same ratio in (x_p, t, Q^2) bins, $\alpha = \phi_e - \phi_S, \beta = \phi_t - \phi_S$

2 · Angles and frames

For an unpolarized electron on a spin-1 target whose alignment axis has polar angle θ_S and azimuth φ relative to the virtual photon and the lepton plane, the Hoodbhoy–Jaffe–Manohar cross section for the projection state m reads [1,2]

$$\frac{d\sigma^{(m)}}{dx dy d\phi} = \frac{2y\alpha^2}{Q^2} \left[F_1 + \frac{2}{3} a_m b_1 + \frac{1-y}{xy^2} \left(F_2 + \frac{2}{3} a_m b_2 \right) - \frac{1-y}{y^2} c_m \sin^2 \theta_S \Delta(x, Q^2) \cos 2\phi \right], \quad c_m = 3m^2 - 2$$

and a fill with populations p_m along a transverse axis ($\theta_S = 90^\circ$) has the φ' distribution $1 + a_2 \cos 2\varphi'$ with

$$a_2 = -P_{zz} \frac{1-y}{y^2} \frac{\Delta}{D_\phi}, \quad D_\phi = F_1 + \frac{1-y}{xy^2} F_2, \quad \hat{A} \equiv \frac{a_2}{P_{zz}}, \quad P_{zz} = \sum_m p_m c_m.$$

Which angle. The φ of the master formula is the azimuth of the transverse spin about q from the lepton plane, in a frame in which P and q are collinear. The covariant definition used for every transverse-spin azimuth in SIDIS and exclusive analyses [3,4] applies verbatim to the spin four-vector S , and with P' in place of S to the recoil azimuth of §4:

$$\cos \phi_S = -\frac{l_\perp \cdot S_\perp}{|l_\perp| |S_\perp|}, \quad \sin \phi_S = -\frac{\epsilon_{\mu\nu\rho\sigma} l^\mu S^\nu P^\rho q^\sigma}{(P \cdot q) \sqrt{1 + \gamma^2} |l_\perp| |S_\perp|}, \quad v_\perp^\mu = g_\perp^{\mu\nu} v_\nu, \quad \gamma^2 = \frac{M^2 Q^2}{(P \cdot q)^2}$$

with $g_\perp$ the projector orthogonal to P and q [3]. In the head-on frame — ion along $+z$, electron along $-z$, the frame of the whole fast simulation — with the spin axis transverse to the beam at lab azimuth φ_S , this reduces *exactly* to $\varphi' = \varphi_e - \varphi_S$ for a massless target and up to $O(\gamma^2)$ for the nucleus: below 5 mrad for $\gamma^2 \leq 0.02$, i.e. everywhere in the sweet spots (`reco.azimuth_wrt_lepton_plane`, verified against an explicit collinear-frame construction to 10^{-13}). The chain uses the covariant formula throughout, since the coherent channel needs it.

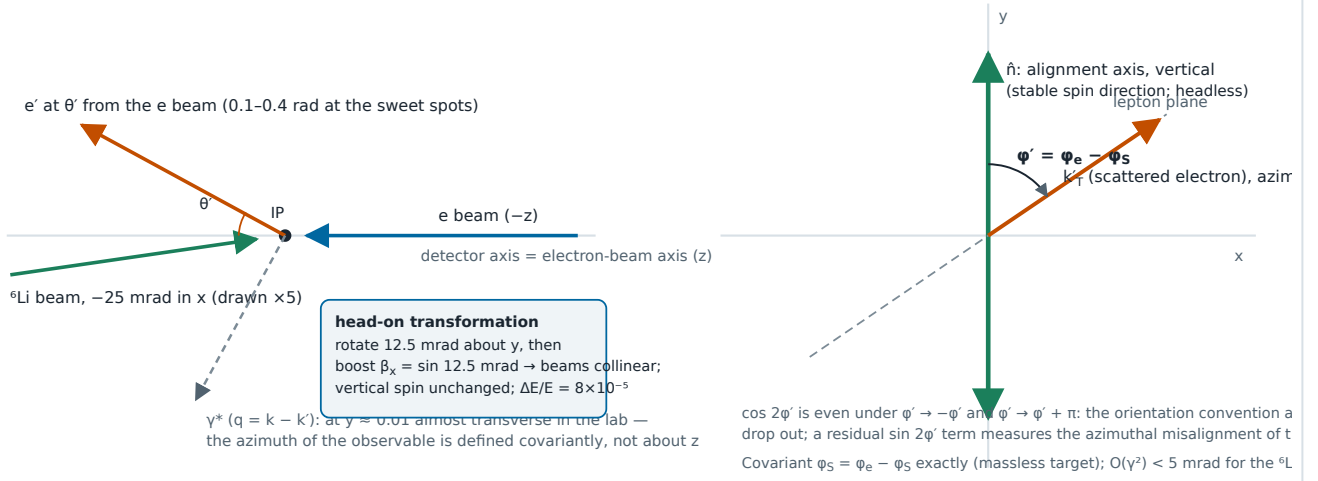
Which frame. The beams cross at 25 mrad in the horizontal plane at IP6; the ePIC axis is the electron-beam axis and the ion beam enters at -25 mrad in x . The head-on frame is a rotation of 12.5 mrad about y and a boost $\beta_x = \sin 12.5$ mrad: energies change by 8×10^{-5} , the vertical spin direction not at all. For the electron the two cancel at first order, leaving an odd-harmonic azimuthal distortion of 0.7×10^{-3} ($\theta' = 107$ mrad) to 2.4×10^{-3} (383 mrad) in $\cos \varphi$, $\cos 2\varphi$ untouched to 10^{-17} and the response of a true $\cos 2\varphi$ term one within 10^{-5} — no fake $\cos 2\varphi'$ for the electron. The recoil is another matter (§4).

Orientation and the axis. $\cos 2\varphi'$ is even under $\varphi' \rightarrow -\varphi'$ and $\varphi' \rightarrow \varphi' + \pi$, and the axis is headless ($m = +1$ and $m = -1$ enter through $c_m = 3m^2 - 2$ symmetrically), so the orientation convention and the sign of $\hat{n}$ drop out; only the direction modulo π matters. A misalignment δ turns the modulation into $\cos 2\delta \cos 2\varphi' + \sin 2\delta \sin 2\varphi'$, and since $\cos 2\varphi'$ is the only physics harmonic in the tensor ratio (m -symmetric fills set $a_1 = 0$; the b_1 -type constant is φ' -independent), a fitted $\sin 2\varphi'$ term measures δ from the data themselves. A polar tilt ε out of the transverse plane dilutes by $\sin^2 \theta_S = 1 - \varepsilon^2$ (0.8% at 5°).

Harmonics at finite γ . The exact spin-1 decomposition (Cosyn et al. [12], Eq. 10) adds tensor pieces with the same azimuthal signatures: $b_2, b_3, b_4 \times \cos \varphi_{TL}$ at order γ (Eq. 17d), $b_2, b_3 \times \cos 2\varphi_{TT}$ at order γ^2 (Eq. 17e), $\gamma = 2Mx/Q$. At the sweet spots $\gamma^2 \leq 0.023$, and the leakage into $\cos 2\varphi'$ is bounded by $\gamma^2(b_1/F_1) \times 1.15$ — the full Eq. (17e) combination, $\approx 6.9 \times$ the $\gamma^2(b_1/F_1)/6$ of the b_1 term alone (plans/08 D2, correcting code review G2) — so with HERMES' $b_1 \approx 0.11$ at $x = 0.012$, i.e. $b_1/F_1 \approx 0.01$ – 0.02 , it is $\leq 0.15\%$ of everything published here, and computable in any case from the same data's A_{zz} ; the $O(\gamma) \cos \varphi'$ term is orthogonal to the fit at full coverage. For the axis along q the T_{LT} and T_{TT} parameters vanish (Eq. 20), which is why the transverse axis is the only choice for Δ . At $x \gtrsim 0.2$ and $Q^2 \approx 1$ GeV² ($\gamma^2 \approx 0.1$) the leakage exceeds a $\Delta/F_1 \sim 10^{-3}$ scenario outright and the A_{zz} subtraction is required (code review G2).

Lab (detector) frame — side view, horizontal x-z plane

Head-on frame — transverse plane, viewed along the ^6Li



Schematic 1 — frames and the azimuth of the observable. Left: the detector frame with the 25 mrad crossing angle, the electron measured at (E', θ', ϕ_e) about the electron-beam axis, the virtual photon far from that axis at low y . Right: in the head-on frame the alignment axis is vertical and the physics azimuth is the angle between the lepton plane and the axis, which the covariant formula reproduces as $\phi_e - \phi_S$.

3 • The inclusive chain

- 1. Event selection and electron identification.** One e' candidate, track matched to an EMCal cluster; E/p and shower-shape cuts against photoproduced π^- , and a charge-symmetric subtraction for the pair-symmetric background at high y . $\epsilon_{\text{eID}}(\eta) = 0.70\text{--}0.95$ on the ATHENA/ECCE anchors of the money_delta study; no official ePIC curve exists.
- 2. Scattered-electron kinematics.** E' from the calorimeter ($2\%/\sqrt{E} \oplus 1\%$, 1.2% at $E' \approx E_e$) or the tracker (3% at $\eta \approx -3$ in the tracking-only table), θ' and ϕ_e from the track at the vertex ($1\text{--}3 \text{ mrad}$), transformed to the head-on frame; k, k', P and $S = (0, \hat{n})$ enter the covariant azimuth.
- 3. Kinematic reconstruction.** The electron method,

$$Q_e^2 = 4E_e E' \sin^2 \frac{\theta'}{2}, \quad y_e = 1 - \frac{E'}{E_e} \cos^2 \frac{\theta'}{2}, \quad x_e = \frac{Q_e^2}{s y_e}, \quad \frac{\delta y_e}{y_e} = \frac{1-y}{y} \left[\frac{\delta E'}{E'} \oplus \tan \frac{\theta'}{2} \delta \theta' \oplus \frac{\delta E_e}{E_e} \right]$$

has a good Q^2 ($\delta Q^2/Q^2 \approx \delta E'/E' \oplus \cot(\theta'/2) \delta \theta'$, angle-dominated at the sweet spots: 5.2% for the 3 mrad track table of §7 at $\theta' \approx 0.1 \text{ rad}$, 1.6% at spot 4, 2% if the backward disks reach 1 mrad — that table is an unsourced placeholder) but loses y as $(1-y)/y$: at $y = 0.01$ a 1.2% energy resolution gives $\delta y/y = 120\%$, the electron-ring energy spread alone ($\approx 10^{-3}$) 10% . Table 1 places money plot 5: at the mid energy three of the four spots are at $y = 0.010\text{--}0.025$, the electron carrying $98\text{--}99\%$ of the beam energy at $\eta \approx -2.9$ to -2.4 , and at the top energy all four sit at $y \approx 0.01$. The remedy is the one HERA used at low y [5,6]: y from the hadronic final state,

$$y_{JB} = \frac{\Sigma}{2E_e}, \quad y_{\Sigma} = \frac{\Sigma}{\Sigma + E'(1 - \cos \theta_e)}, \quad \Sigma = \sum_{h \neq e'} (E_h - p_{z,h}), \quad x = \frac{Q_e^2}{s y_{\Sigma}} \text{ (mixed method)}$$

with θ_e measured from the ion direction. The Σ method is insensitive to collinear initial-state radiation and both methods to particles lost along the ion beam ($E - p_z \approx 0$ for forward remnants) — but the insensitivity is to y alone: the mixed $x = Q_e^2/(s y_{\Sigma})$ takes its numerator from the electron, so a photon radiated before the vertex carries $x \rightarrow x/(1-z)$ exactly, and the variable the analysis bins in is not ISR-robust even though the y it is built from is (plans/07 WP4); their resolution is set by hadronic calorimetry and by losses in the electron direction. Three documented expectations bracket it: ATHENA, $e\text{--}\Sigma$ or DA for $y \lesssim 0.1$, $\delta y/y \approx 25\%$ at $y \approx 0.01$ and $\approx 10\%$ at $y \approx 0.1$, JB $20\text{--}30\%$ throughout (Sec. 3.1, Fig. 22) [18]; the ePIC kinematic-fit study, hadronic $E - p_z$ smeared by $\sigma(\delta_h)/\delta_h = 25\%$, $\Delta y/y$ widths $0.2\text{--}0.3$ in $0.01 < y < 0.05$ [19]; the ATHENA fast simulation of Arratia et al., $\text{RMS}(y)/y \approx 0.13\text{--}0.17$ for $\text{I}\Sigma$, DA and JB at $y \approx 0.05\text{--}0.2$ [20]. The band assumed here is $15\text{--}30\%$, 25% central (the value used in §7). The low-energy configuration moves the same (x, Q^2) to $y = 0.05\text{--}0.25$ (Table 1), so the energy scan is part of the reconstruction strategy for the $x \approx 0.1$ bins.

SWEET SPOT	X, Q ² [GeV ²]	E(5)× ⁶ Li(20.5): Y	E(10)× ⁶ Li(50): Y, E' [GeV], Θ' [MRAD], H	E(18)× ⁶ Li(137.5): Y	ΔY/Y, E' ALONE (1.2%)
1	0.056, 1.14	0.049	0.010, 9.93, 107, -2.93	0.011 (x = 0.014)	1.2
2	0.022, 1.14	0.124	0.025, 9.77, 108, -2.92	0.026 (x = 0.0056)	0.46
3	0.141, 3.13	0.054	0.011, 9.97, 177, -2.42	0.011 (x = 0.028)	1.1
4	0.141, 14.3	0.247	0.051, 9.85, 383, -1.64	0.010	0.23

Table 1 — the four sweet-spot super-bins of money plot 5 (selected per Q² band from the significance map at e(10)×⁶Li(50)) and where the same (x, Q²) bins sit at the low energy (y = Q²/(sx), s = 4·5·20.5 GeV²); the top-energy spots follow the same rule from the top-energy map and land at different x. Figure 1(b) places all three energies.

The hadronic final state in the chain. `polligen.hfs` carries the hadronic final state of unpolarized DIS events through a hadron-side response and the Σ , Jacquet–Blondel and double-angle methods, as a library for the pseudo-experiments: the captured fraction of $\Sigma = \Sigma(E - p_z)$ and the p_T ratio of a generator event from the same (x, Q²) cell are applied to the pseudo-event's exact truth, noise is added per event, and y follows with the *reconstructed* electron. The tensor polarization enters only the azimuthal weight, so an unpolarized generator is the right source: PYTHIA 8, 12 M events over the three beam configurations, e + p and e + n at the per-nucleon energies with lepton radiation off (`tools/pythia8/gen_dis_hfs.py` ; manifest in `evgen/samples/README.md`), with `hfs.ToyHFS` as a labelled fallback. The response: tracks within $|\eta| \leq 3.5$ above $p_T = 0.2$ GeV (95% plateau), calorimeter objects within $|\eta| \leq 3.7$ (EMCal photons above 0.1 GeV, neutral and untracked charged hadrons in the HCal above 0.5 GeV at 90%), Yellow Report resolutions per region, and Gaussian noise on Σ and on each p_T component — 50 MeV by default, the magnitude Arratia et al. needed to reproduce the H1 full simulation at low y [20]. Figure 3: 0.90 of Σ is captured (0.51 tracks, 0.28 EMCal photons and electrons, 0.11 neutral hadrons, the rest below threshold or beyond $|\eta| = 3.7$); without noise the Σ method resolves y to 18–20% throughout $0.005 < y < 0.05$, and with 50 MeV of noise the four sweet spots give 0.32 / 0.22 / 0.28 / 0.11 (Σ and JB alike; DA 0.49 / 0.37 / 0.40 / 0.15; the electron method 1.17 / 0.46 / 1.05 / 0.22). The noise floor, not the fragmentation, decides the lowest bins: at y = 0.005 the Σ resolution runs 0.20 / 0.32 / 0.54 / 1.01 for 0 / 25 / 50 / 100 MeV of noise and at y = 0.0125 0.18 / 0.21 / 0.27 / 0.44, so the ePIC noise and threshold floor at $\Sigma_h \approx 0.2$ –0.5 GeV sets the y ≈ 0.01 bins (plans/04 #21). It degrades monotonically with beam energy — 0.21 / 0.12 / 0.17 / 0.10 low against 0.74 / 0.34 / 0.69 / 0.18 top, the spots moving to lower y as s grows — so the x ≈ 0.1 bins belong to the low-energy configuration (`hfs_resolution.py --config 0/1/2`).

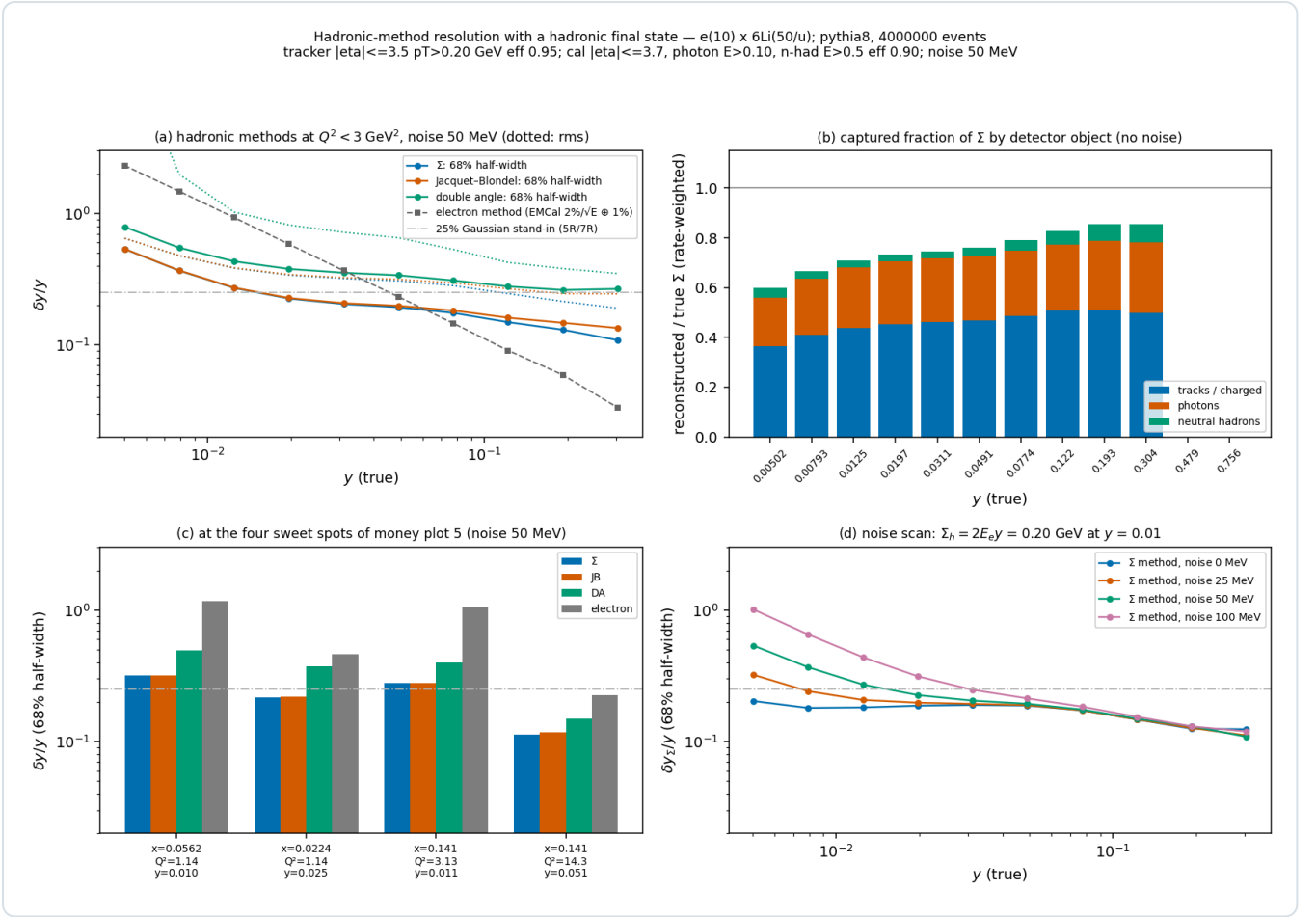


Figure 3 — hadronic-method resolution with a hadronic final state (e10 $\times$ $^6\text{Li}50$, **PYTHIA 8**, 4 M e + p and e + n events, `hfs_resolution.py`). **(a)** $\delta y/y$ (68% half-width; dotted: rms) of the Σ , Jacquet-Blondel and double-angle methods at $Q^2 < 3 \text{ GeV}^2$ with the default response and 50 MeV noise, against the electron method and the 25% stand-in. **(b)** Captured fraction of Σ by detector object, no noise. **(c)** The four sweet spots. **(d)** The Σ -method resolution for 0, 25, 50 and 100 MeV of calorimeter noise.

4. **The azimuth.** $\phi' = \phi_e - \phi_S$ from the covariant formula (§2), ϕ_S being the alignment-axis azimuth of that bunch. The track direction resolution enters as $\sigma_\phi = \sigma_\theta / \sin \theta' \approx 20\text{--}30 \text{ mrad}$ at the sweet spots and dilutes $\cos 2\phi'$ by $\langle \cos 2\delta\phi' \rangle = 0.985\text{--}0.999$: negligible.
5. **Binning in reconstructed (x , Q^2).** Figure 1(c): with $\delta y_{\text{had}}/y = 15\%$ (20%, 30%) the 3×3 super-bins keep a purity of 0.81–0.83 (0.74–0.77, 0.61–0.65) and the amplitude seen in the reconstructed bin is 0.98–0.99 (0.96–0.99, 0.92–0.98) of the true-bin value, because $A(x)$ varies slowly across the bin; with the electron method alone the low- y spots collapse to purities of 0.41–0.56 with 26–53% of the events surviving. The correction is the standard migration matrix, built on the hadronic-method resolution.
6. **Spin-state sorting and the estimator.** The events of each (x, Q^2) bin are histogrammed in ϕ' per spin state f , $N_f(\phi')$. Any smooth $\epsilon(\phi')$ multiplies every N_f identically, so the bin-wise ratio

$$R(\phi') = \frac{\sum_f w_f N_f(\phi')}{\sum_f N_f(\phi')}, \quad w_f = P_f - \bar{P}, \quad \bar{P} = \frac{\sum_f L_f P_f}{\sum_f L_f}, \quad R = \frac{\sigma_P^2 T(\phi')}{1 + \bar{P} T(\phi')}, \quad T = \kappa + \hat{A} \cos 2\phi', \quad \sigma_P^2 = \frac{\sum_f L_f (P_f - \bar{P})^2}{\sum_f L_f}$$

is free of $\epsilon(\phi')$ exactly (raw counts with luminosity-weighted w_f , so that $\sum_f L_f w_f = 0$), and a relative-luminosity error enters only through $\bar{P}$, as a ϕ' -independent constant plus a second-order rescaling ($\delta \times \bar{P}/\sigma_P^2$). It presumes both spin states see the *same* $\epsilon(\phi')$ — the one systematic the ratio cannot cancel, and the reason the states must alternate bunch by bunch within a fill or the harmonic be stable to 10^{-4} between fills (§7.1). A weighted least-squares fit of $T(\phi') = c + d \cos 2\phi' (+ e \sin 2\phi')$, T obtained from R bin by bin by an exact inversion, gives $\hat{A} = d$ once the finite-bin dilution $\sin w/w$ is divided out; $\kappa = c$ is the transverse tensor asymmetry of the b_1 sector, a by-product whose sign follows the program's b_1 transcription rather than

the HJM/HERMES convention (Cosyn et al. [12], Eq. 27: $A_{zz} = -(2/3)b_1/F_1$; code review G1) and changes nothing in the Δ sector. For two fill types of equal luminosity the statistical error is

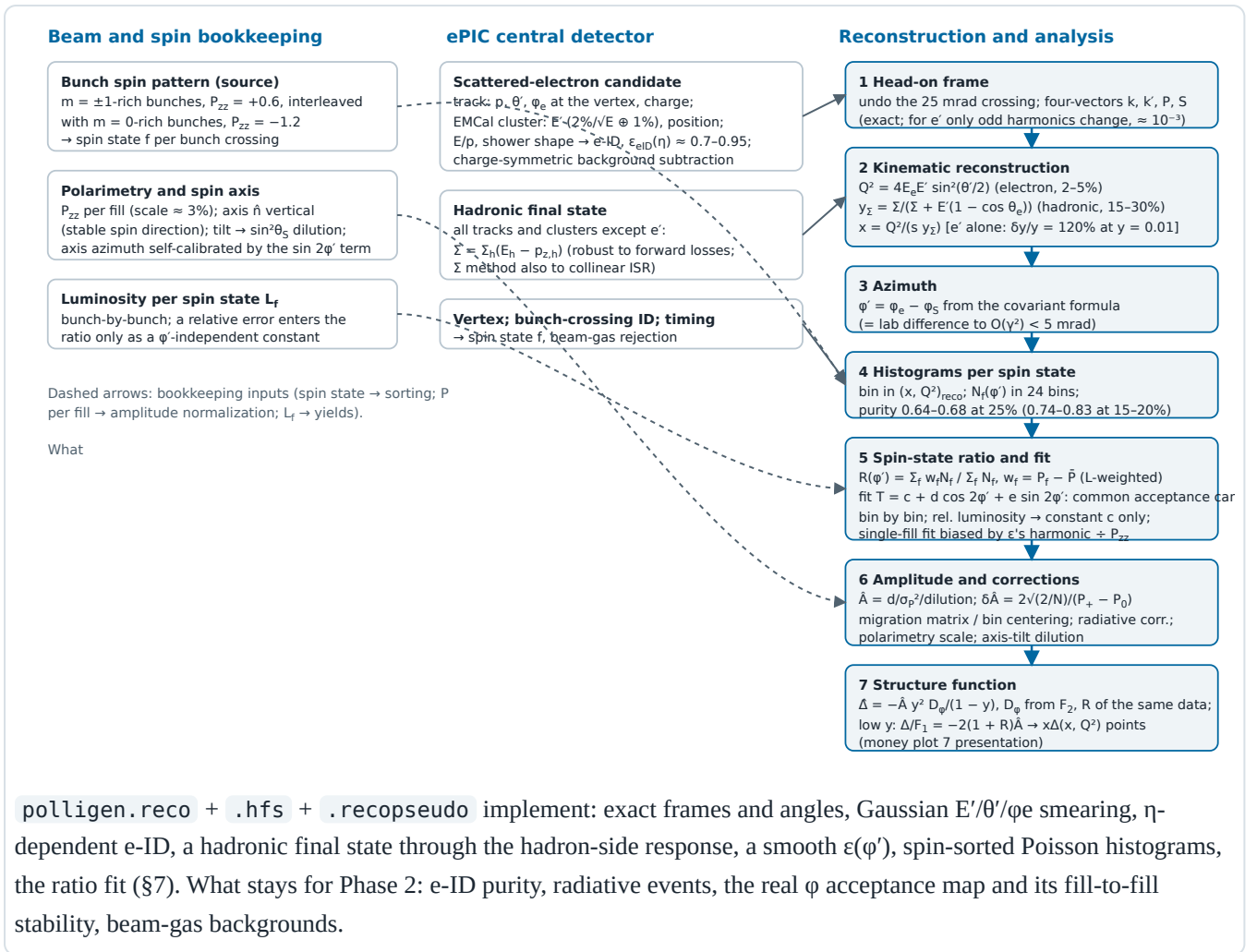
$$\delta\hat{A} = \frac{2}{P_+ - P_0} \sqrt{\frac{2}{N}} \quad (\text{two fills}) \quad \text{vs} \quad \delta\hat{A} = \frac{1}{P_{zz}} \sqrt{\frac{2}{N}} \quad (\text{single fill})$$

so $m = \pm 1$ -rich bunches at $P_+ = +0.6$ alternated with $m = 0$ -rich bunches at $P_0 = -1.2$ (the same source purity gives $-2P_+$, the pattern of the A_{zz} run plan) are $1.5\times$ more precise than a single $P_{zz} = 0.6$ fill with the whole luminosity; an unpolarized reference fill ($P_0 = 0$) would cost a factor two instead. Figure 1(d): under $\varepsilon = 1 + 0.03 \cos 2\varphi' + 0.02 \cos \varphi'$ the single-fill fit returns $\hat{A} = 0.062$ for a true 0.0118 — the acceptance harmonic divided by P_{zz} — while the ratio is unbiased with the analytic error. A percent-level $\cos 2\varphi$ harmonic aligned with the vertical axis is to be expected: the crossing-angle plane is horizontal, module boundaries and dead areas are not azimuthally random, and the Roman-Pot cutout is a rectangle.

7. **Amplitude to Δ .** With F_2 and $R = \sigma_L/\sigma_T$ of ${}^6\text{Li}$ per nucleon measured in the same data (the unpolarized cross section of the same sample, with its own radiative corrections) or taken from nuclear-PDF fits,

$$\hat{\Delta}(x, Q^2) = -\hat{A} \frac{y^2 D_\phi}{1-y} \rightarrow -\hat{A} \frac{F_2}{x} = -2(1+R) F_1 \hat{A} \quad (y \ll 1), \quad \text{i. e.} \quad \frac{\Delta}{F_1} = -2(1+R)\hat{A}$$

At the sweet spots the conversion is insensitive to y — the $(1-y)/y^2$ factor that makes low y attractive for the amplitude cancels against D_ϕ — so Δ/F_1 is essentially a direct ratio measurement, and the y mis-measurement of the electron method moves the x label of the point, not the amplitude. Of the two structure-function inputs only R moves the answer: F_2 cancels *exactly* in the $\cos 2\varphi$ amplitude (2×10^{-16} residual, two independent levers). The chain takes R from the published SLAC/E143 R1998 world fit [21]; against the simplified parametrization it replaces, Δ/F_1 moves $+16.6 / +18.0 / +4.7 / -4.4\%$ at the four sweet spots. Remaining corrections: the tensor rate shift $(2/3)a_m(b_1 + \dots)$ in D_ϕ , at the 10^{-3} level and measured by κ ; target-mass terms of order y^2 ; radiative corrections, uncalculated for tensor observables (plans/07 WP4); the per-nucleon and dilution conventions, which are interpretation rather than reconstruction; and the polarimetry scale $\delta P_{zz}/P_{zz} \approx 3\%$, common with A_{zz} .



`polligen.reco` + `.hfs` + `.recopseudo` implement: exact frames and angles, Gaussian $E'/\theta'/\phi_e$ smearing, η -dependent e^-ID , a hadronic final state through the hadron-side response, a smooth $\epsilon(\phi')$, spin-sorted Poisson histograms, the ratio fit (§7). What stays for Phase 2: e^-ID purity, radiative events, the real ϕ acceptance map and its fill-to-fill stability, beam-gas backgrounds.

Schematic 2 — the inclusive reconstruction chain. Left and middle: what is recorded; right: the seven reconstruction and analysis steps of §3 with the quantity each produces and its dominant limitation. The observable is the $\cos 2\phi'$ coefficient of the spin-state ratio (step 5), not the modulation of a single histogram.

Inclusive $\cos 2\phi$ measurement: what the scattered electron can and cannot deliver, and what the estimator must cancel

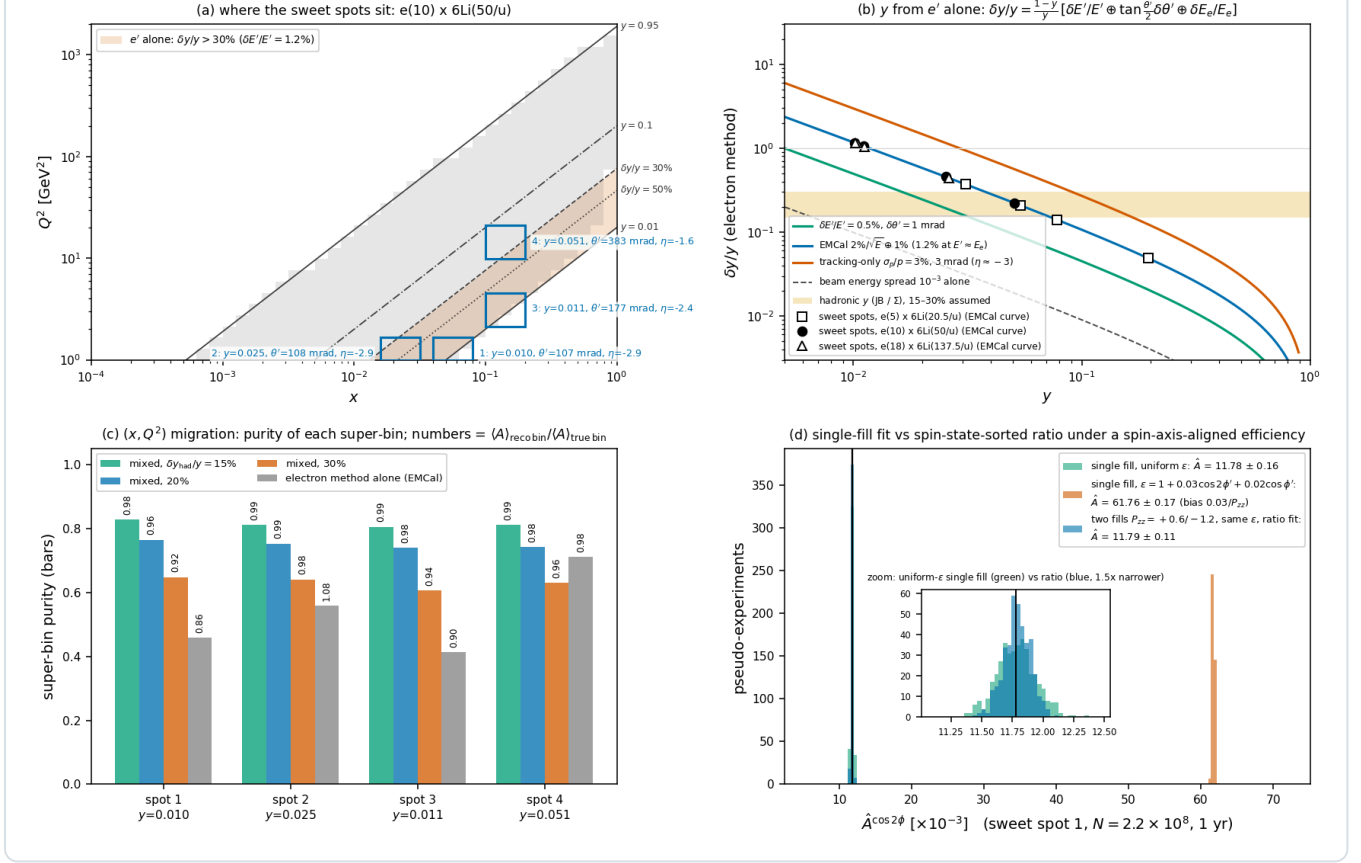


Figure 1 — the inclusive measurement quantified ($e10 \times 6\text{Li}50$, `reco_chain_figures.py`). **(a)** The analysis acceptance with iso- y lines and the four sweet-spot super-bins; the shaded band is where the electron alone gives $\delta y/y > 30\%$ at a 1.2% energy resolution — spots 1–3 inside it, spot 4 just above. **(b)** The electron-method $\delta y/y$ versus y for calorimeter, tracking and beam-energy-spread inputs, with the sweet spots of the three energies and the assumed 15–30% hadronic band. **(c)** Purity of each super-bin for three hadronic- y resolutions and for the electron method alone; the numbers give the amplitude averaged over the events landing in the reconstructed bin over that of the true bin. **(d)** Four hundred pseudo-experiments of the spot-1 super-bin at 1-year statistics: single-fill fit with a uniform efficiency (green), the same under $\epsilon = 1 + 0.03 \cos 2\phi' + 0.02 \cos \phi'$ (vermillion, biased by $0.03/P_{zz}$) and the two-fill ratio fit under that efficiency (blue, unbiased and 1.5× more precise; inset).

4 • The coherent chain

The coherent measurement keeps steps 1–7 for the electron and adds the recoil. Two facts fix its kinematics. The recoil keeps the beam rigidity ($R = 1.000$ for $A/Z = 2$, and $x_p < 0.01$ is below the Roman-Pot momentum resolution), so the only measured recoil quantity is its transverse displacement from the beam orbit; and coherence confines the sample to $x \lesssim 0.01$, i.e. $y \gtrsim 0.05$ at the mid energy (median $x = 3.5 \times 10^{-3}$, $y \approx 0.2$), where the low- y problem of §3 does not arise.

1. Roman-Pot track. Hits in the two AC-LGAD stations ($\approx 30 \text{ ps}$ timing for bunch association) form a local track, which the optics transfer matrices referenced to the beam orbit turn into the IP angle pair (θ_x, θ_y) relative to the ion axis; crossing angle and reference orbit drop out by construction, the central-detector vertex supplying the position term. x_L is unresolved and set to 1, so $p_T = A p_u \sqrt{(\theta_x^2 + \theta_y^2)}$ and $\phi_t = \text{atan2}(\theta_y, \theta_x)$. The bunch's own divergence dominates the resolution: $\delta p_T = A p_u \sigma_\theta = 22 \text{ MeV}$ at 50 GeV/u for $\sigma_\theta = 73 \text{ } \mu\text{rad}$, hence $\sigma(\phi_t) = 0.09 \text{ rad}$ at $p_T = 0.25 \text{ GeV}$ and a $\cos 2\phi_t$ dilution of 1.5% (Figure 2d). The ePIC far-forward simulation with all beam effects gives $\Delta p_T \approx 40 \text{ MeV}$ for 275 GeV protons ($\Delta p_T/p_T \approx 0.09$ at 0.45 GeV , 0.04 at 1 GeV), the detector alone $\leq 1.5\%$ — "beam effects the dominant source of momentum smearing" [17]; ZEUS's leading-proton spectrometer was limited the same way ($\sigma(t)/t = 0.14 \text{ GeV}/\sqrt{|t|}$, Φ resolution 0.2 rad) [13].

2. Momentum transfer. Exactly, with $x_L = 1 - x_p$ the plus-momentum fraction,

$$t = -\frac{p_T^2 + M^2 x_p^2}{1 - x_p} \simeq -p_T^2, \quad t_{\min} = \frac{M^2 x_p^2}{1 - x_p} = 3.2 \times 10^{-3} \text{ GeV}^2 \quad (x_p = 0.01), \quad x_p = \frac{M_X^2 + Q^2 - t}{W^2 + Q^2 - M^2}$$

so the Roman Pots give t , while x_p (or $\beta = x_A/x_p$) must come from the diffractive mass M_X of the central detector (the HERA M_X method [16]) — a coarse variable, a few bins, enough for the x_p -dependence test of the two-component fit. The covariant recoil azimuth differs from $\varphi_e - \varphi_t$ by less than 0.2 mrad here.

3. Exclusivity selection. A beam-rigidity Roman-Pot track above the envelope; no OMD track (p at $R = 0.5$, ^3He at 0.75 in the main window [8]), no ZDC neutron, no B0 EMCal photon (the 3.56 MeV state's de-excitation γ , boosted to 0.1–0.4 GeV); a rapidity gap; Z-identification if the pots resolve $Z = 3$ from 2 and 1 (open question #19), otherwise a two-component fit of the $|t|$ spectrum against the ten-times-flatter $\alpha + d$ breakup slope; timing and vertex association against beam-gas coincidences. The incoherent cross section is m -state-blind, so residual breakup dilutes the tensor ratio without faking it, and its fraction comes from the same $|t|$ fit.

4. Three azimuths, two harmonics. With the recoil measured there are two physical azimuths besides the spin: $\alpha = \varphi_e - \varphi_S$ and $\beta = \varphi_t - \varphi_S$, with $\alpha - \beta = \varphi_e - \varphi_t$ the lepton-plane–recoil angle of unpolarized diffractive DIS. The rank-2 polarization enters only through $e^{\pm 2i\varphi_S}$, so at leading power in the momentum transfer the spin-sorted distribution is

$$N_f(\alpha, \beta) \propto \varepsilon(\alpha, \beta) [1 + u_1 \cos(\alpha - \beta) + u_2 \cos 2(\alpha - \beta) + P_f(\kappa + a_e \cos 2\alpha + a_t \cos 2\beta + a_m \cos(\alpha + \beta))]$$

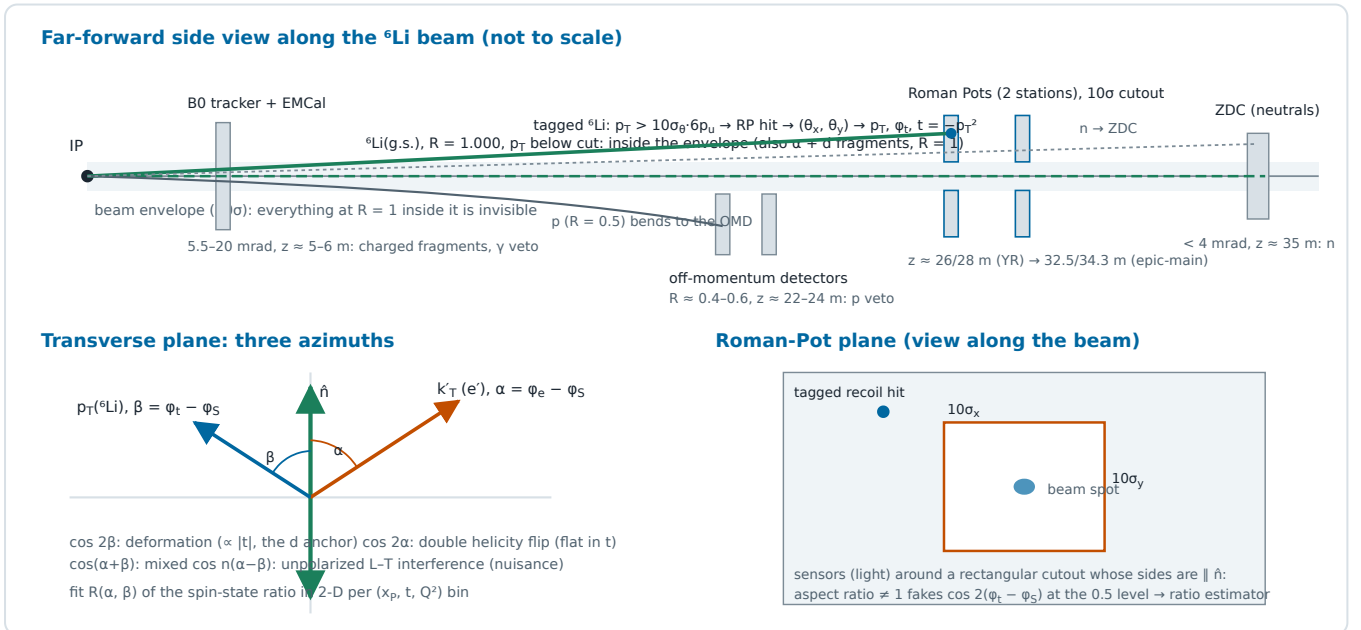
The deformation term of the polarized-deuteron anchor — the diffractive slope modulated by the aligned gluon density, $a_2(t) \propto |t|$, digitized in `coherent.MANTYSAARI_A2_DEUTERON` — is a_t : it lives in the *recoil* azimuth β , needs no photon helicity flip, and vanishes as $t \rightarrow 0$. (The anchor's Φ is defined in the γ^*d c.m. frame, where at $Q^2 > 0$ the transformation from the lab "would mix the polarization states" [9] — the $O(\gamma)$ mixing the covariant definitions of §2 absorb.) The double-helicity-flip term, the coherent counterpart of Δ , needs the photon's linear polarization and hence the lepton plane: it is a_e , lives in α , and is flat in t at leading power. The two mechanisms are therefore separated by the azimuth they modulate as well as by their t -dependence; a_m and the unpolarized $u_{1,2}$ are nuisances. The latter are the L–T and T–T' interference terms of diffractive DIS with a detected recoil, $u_1 = (2-y)\sqrt{(1-y)} A_{LT}/[2(1-y)+y^2]$ and $u_2 = 2(1-y) A_{TT}/[2(1-y)+y^2]$ [14]; ZEUS measured $A_{LT} = -0.036 \pm 0.036$, $A_{TT} = -0.030 \pm 0.037$ for $x_p < 0.01$ and $+0.051 \pm 0.024$, -0.010 ± 0.024 for $0.01 < x_p < 0.1$ [13], and the pQCD LT-interference model gives $A_{LT} \approx 0.03$ at our $\beta \approx 0.5$, $p_T/Q \approx 0.2$ [15]; §7 uses $u_1 = 0.05$, $u_2 = 0.02$, at the edge of those bounds. The measurement histograms (α, β) in two dimensions per (x_p, t, Q^2) bin and fits a_t and a_e from the spin-state ratio $R(\alpha, \beta)$ — money plot 6 places the anchored $\langle a_2 \rangle$ on the electron axis α , harmless for a statistical projection — with $u(\alpha - \beta)$ carried, since it survives in the ratio's denominator. An *error* δu does not stay second order: it reaches the pure harmonics as $a_t \cdot \delta u_2 \cdot \langle \cos 2\beta \rangle$ and $a_e \cdot \delta u_1$, and with $|\langle \cos 2\beta \rangle| \approx 0.77$ from the cutout and $a_t/a_e \approx 12$ –34 the first dominates: a ZEUS- 1σ $\delta u_2 = 0.024$ shifts a_e by 20%, 1.6 one-year statistical errors, while the same δu_1 moves it by under 1%. u_2 , not u_1 , is the nuisance to measure in situ from the spin-averaged counts.

The forbidden harmonics. With unpolarized electrons and a headless axis the cross section is even under $(\alpha, \beta) \rightarrow (-\alpha, -\beta)$, so $\sin 2\alpha$, $\sin 2\beta$ and $\sin(\alpha + \beta)$ vanish exactly: the coherent null test, counterpart of the inclusive $\sin 2\varphi'$ term. Their pattern says *which* azimuth is misaligned — a spin-axis error δ gives $\sin/\cos = \tan 2\delta$ in α and β , a Roman-Pot azimuthal roll δ_t gives $\tan 2\delta_t$ in β alone and leaves the lepton-plane harmonic exactly null (verified to 10^{-5} in `tests/test_recopseudo.py`). The recoil null is the sensitive one, resolving a pot roll to 5–8 mrad per $|t|$ bin in a year against the α null's 60–300 mrad on the spin axis. The sin columns are orthogonal to the cos ones on the full grid, so the null costs nothing; its t -dependent sine must use the same t -shape template as its cosine partner, since with a plain $\langle \sin 2\beta \rangle$ it misses its own closure by 5%.

5. Acceptance geometry and the estimator. The acceptance is not a circular p_T cut: the pots are 25.6×12.8 cm AC-LGAD planes surrounding a *horizontal slot* through which the beam, its momentum spread and its dispersion pass [17], and its symmetry axes are horizontal and vertical — parallel and perpendicular to the spin axis. Anisotropy is the norm: HERA's proton beam had a transverse-momentum spread at the IP of 45 MeV horizontally against 100 MeV vertically [13]. For the exponential recoil spectrum the tagged azimuthal distribution is $\varepsilon(\varphi_t) = \exp[-B \rho(\varphi_t)^2]$ with p

the cutout boundary, and its $\langle \cos 2\phi_t \rangle$ about the horizontal runs from 0 for a square through +0.49 at aspect ratio $c_y/c_x = 1.25$ and +0.71 at 1.5 to -0.36 to -0.45 for the slot-like ratios 0.3–0.6 of Figure 2b and -0.77 for the wider slot of §7 (0.55×0.22 GeV) — ten to twenty times the anchored $a_t \approx 0.036$, about the same symmetry axis (the sign flips when the axis is referred to the vertical spin direction). A square still carries $\langle \cos 4\phi_t \rangle = 0.34$, which mixes into $\cos 2\beta$ at second order through the physics term. No single-fill fit survives this; the spin-state ratio of §3 cancels it exactly, bin by bin in (α, β) .

6. **The near-beam cut is angular, and the aperture is measured.** The documented 0.20 GeV (high-acceptance) and ≈ 0.41 – 0.45 GeV (high-divergence) cuts are ten beam divergences times the 275 GeV proton momentum [7], $\sigma_\theta \approx 73$ and $149 \mu\text{rad}$. The same envelope on a ${}^6\text{Li}$ of momentum $6p_u$ is $10\sigma_\theta \cdot 6p_u = 0.09, 0.22$ and 0.60 GeV at 20.5, 50 and 137.5 GeV/u — for circular tag acceptances of 67%, 9% and 1.5×10^{-8} at $B = 50 \text{ GeV}^{-2}$ (Figure 2c), against the projection's constant 0.20 GeV cut (13.5%). The A_{p_u} scaling is kinematics; the lithium divergence itself is undocumented (plans/04 #11). Against that envelope the optics scan (`coherent_optics_scan.py`) gives, at the slot cutout, tagged fractions 32% / 3.0% / 4×10^{-5} / 2×10^{-7} at 0.10 / 0.22 / 0.45 / 0.60 GeV and one-year errors on the deformation coefficient of 1.6% / 5.7% / 104% / 540%. The aperture itself is measured, not assumed: an intact ${}^6\text{Li}$ through the ePIC far-forward geometry (`tools/fullsim/ion_gun_hepmc.py` , 84 points in $p_T \times \text{azimuth}$) clears the pots at $|\theta_x| \gtrsim 2.0 / 1.35 / 1.03$ mrad in the $5 \times 41 / 10 \times 100 / 18 \times 275$ optics — $p_T = 0.25 / 0.41 / 0.85$ GeV — against $|\theta_y| \gtrsim 1.8$ – 3 mrad, an α at the same rigidity clearing at the same angle, so this is optics, not species: the beamline images an IP angle onto the pot plane with a horizontal lever of 30.6 m against a vertical few metres. The aperture is tighter than the 10σ envelope at every energy and inverts the assumed aspect, so the fake $\langle \cos 2\beta \rangle$ changes sign; a track must clear both, and the tagged fraction at $B = 50 \text{ GeV}^{-2}$ falls to 1.4×10^{-2} at 20.5 GeV/u, 5×10^{-5} at 50 and 1.9×10^{-17} at 137.5, against $0.82 / 0.40 / 1.6 \times 10^{-2}$ for the slot the chain assumed, whose width in x put it above the circular envelope (§7). The scan is one event per point in 30° azimuthal steps, no beam envelope, September-2024 `epic-main` (plans/04 #20). The alternative is the IR-8 secondary focus, which reaches $p_T \approx 0$ [10] and whose ${}^6\text{Li}$ efficiency is our interpolation of the published $d / {}^3\text{He} / {}^4\text{He} / {}^7\text{Li}$ values.
7. **Resolution and unfolding.** The divergence smearing moves the effective slope from B to $1/(1/B + 2\delta p_T^2)$ (47.7 here) and smears $|t|$ by $2p_T\delta p_T \approx 0.01 \text{ GeV}^2$, small against the 0.04 – 0.2 GeV^2 window (Figure 2d); the t -binning migration and the acceptance edge are unfolded with the same emulation, and $t_{\min}$ is a known bias of $3 \times 10^{-3} \text{ GeV}^2$ at $x_p = 0.01$ once x_p is known from M_X .



Schematic 3 — the coherent channel. Top: the far-forward measurement of the intact recoil, visible only when its transverse momentum carries it outside the beam envelope at the Roman Pots, and the vetoes that define exclusivity. Bottom left: the two physical azimuths relative to the alignment axis and the harmonic each mechanism modulates. Bottom right: the rectangular cutout, whose symmetry axes coincide with the spin axis.

Coherent intact- ${}^6\text{Li}$ channel: the recoil measurement, its acceptance geometry, and the angular near-beam cut

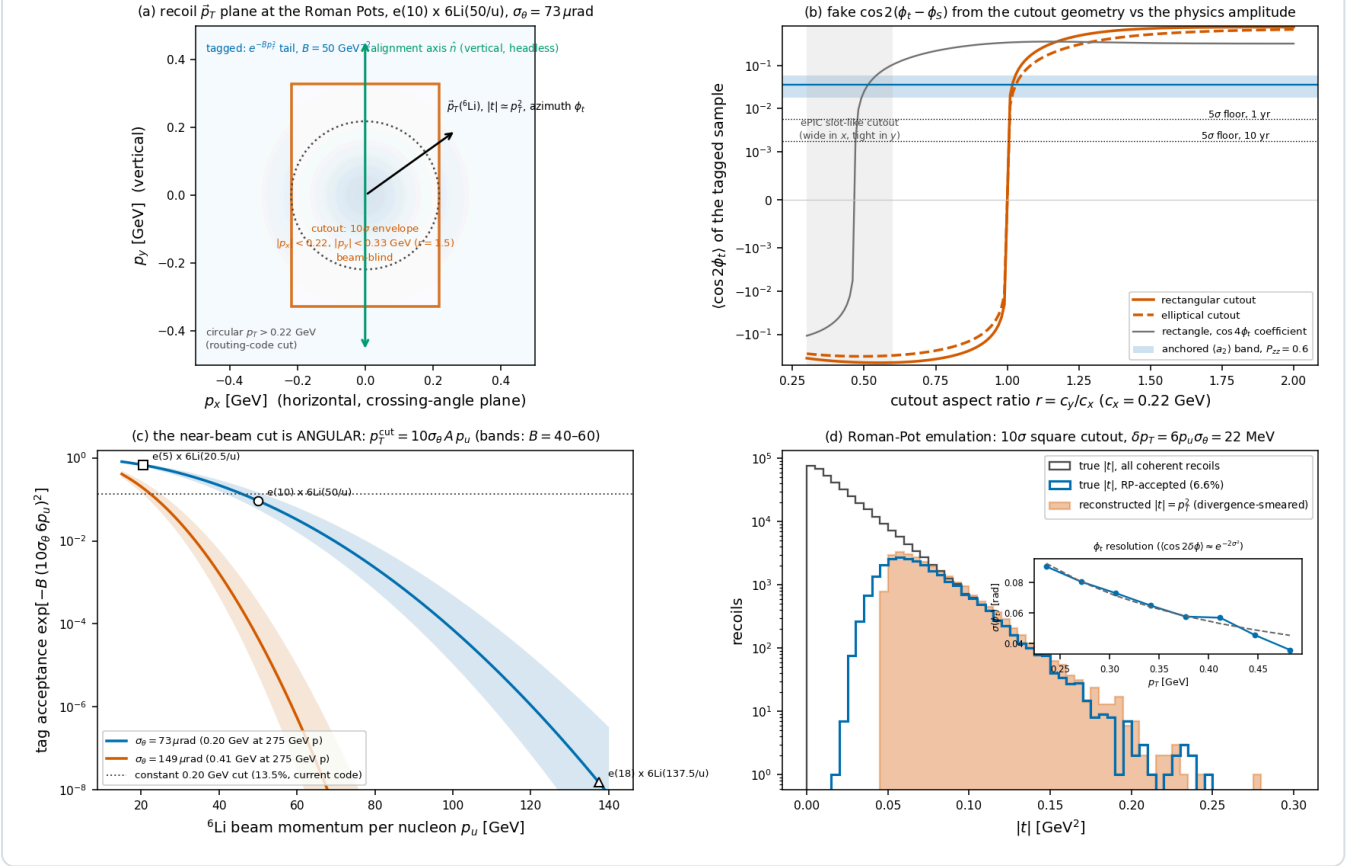


Figure 2 — the coherent measurement quantified. **(a)** The recoil transverse-momentum plane: the exponential coherent spectrum, the rectangular 10σ cutout (aspect ratio 1.5 drawn) against the circular cut of the routing code, and the vertical alignment axis. **(b)** $(\cos 2\phi_t)$ of the tagged sample versus cutout aspect ratio for rectangular and elliptical cutouts, with the $\cos 4\phi_t$ coefficient of the square, the anchored deformation band and the 5σ statistical floors of the 1- and 10-year programs. **(c)** Tag acceptance versus beam momentum per nucleon for the two proton-derived divergences, with the $B = 40\text{--}60 \text{ GeV}^{-2}$ bands, the three beam configurations and the constant-cut value of the projection. **(d)** The Roman-Pot emulation (`reco.rp_measure`): true and accepted $|t|$ spectra, the reconstructed $|t| = p_T^2$ after divergence smearing, and (inset) the ϕ_t resolution.

5 • What the simulation does

Behind the measurement layer sit the generator kernel and the spin bookkeeping (`bookkeeping.SpinCategory` , `RunPlan` , `tensor_flip_plan` : spin categories with luminosity shares, relative-luminosity offsets, polarimetry smearing). The layer itself is `polligen/reco.py` and `polligen/hfs.py` [11] — every step of §§3–4, from the head-on transformation to the Roman-Pot measurement with both the angular envelope and the measured aperture — and `polligen/recopseudo.py` composes it into the pseudo-experiments of §7, where `RecoResponse.fold` is the response as an exact linear operator on Δ .

Four things it does not model, each limiting a number here. The track angular resolution table (1–5 mrad by η) has no published source, so the Q^2 resolution of §3 and the azimuthal dilution rest on a placeholder. F_2 comes from the model rather than from the same data — immaterial for Δ/F_1 , which is F_2 -free (§3), but not for a Δ in absolute units. Radiative corrections are absent, for the yields and the tensor observables alike (plans/07 WP4). And there is no electron side to the coherent fit: x_p is sampled rather than reconstructed from M_X , α is integrated analytically, and the exclusivity vetoes are bookkeeping (`coherent.veto_table`) with no event-level breakup model.

6 • What is left to build

A harmonic basis that survives the strongly anisotropic measured aperture — fewer columns, wider bins, or $|t|$ re-binned inside the window the cutout leaves — and money plot 6R re-derived on it; the event-level $\alpha + d$ breakup behind the two-

component $|t|$ fit, which needs BeAGLE for $A = 6$ (plans/06 carries the budget); and the four omissions of §5. Item by item that is plans/08 §8.4, and the external inputs it waits on are plans/04.

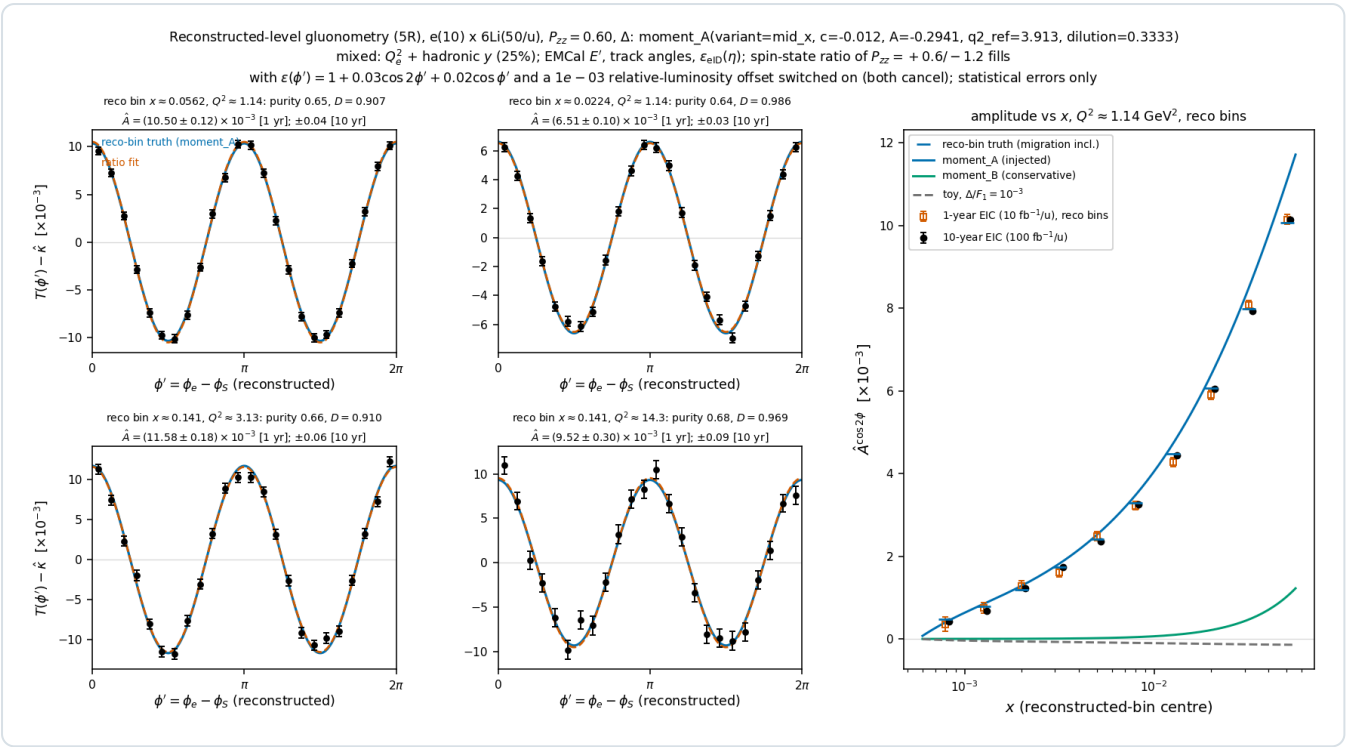
7 • Reconstructed-level closure: money plots 5R, 7R and 6R

The chain of §§3–4 is composed in `polligen/recopseudo.py`. The inclusive response is importance-sampled: every accepted sampler cell (60×45 grid, generator cuts loosened to $y \geq 0.004$, $Q^2 \geq 0.7 \text{ GeV}^2$, $|\eta| \leq 3.8$, $E' \geq 0.3 \text{ GeV}$ so that events can migrate *into* the analysis window) receives 400 pseudo-events of weight σ_c/n_{kept} , n_{kept} counting those inside the generator acceptance so that the cell rate is conserved. Each passes through the lab frame (25 mrad crossing), Gaussian smearing of E' ($2\%/\sqrt{E} \oplus 1\%$) and of the track direction ($1\text{--}5 \text{ mrad}$ by η), the head-on transformation, the electron-method Q^2 , a hadronic y — the documented 25% stand-in [18,19] or, with `--y-source hfs`, the PYTHIA 8 final state through the hadron-side response — the mixed-method x , the analysis cuts on reconstructed quantities, $\epsilon_{\text{eID}}(\eta)$, and the covariant φ' . The expected φ' counts of a reconstructed bin are exact bin integrals of $\text{den}_f + P_f A \cos 2\varphi'$ over the events landing in it, Poisson-fluctuated at the 1- and 10-year luminosities for the two fills of the $m = \pm 1$ -rich / $m = 0$ -rich pattern (`bookkeeping.tensor_flip_plan`), with $\epsilon(\varphi') = 1 + 0.03 \cos 2\varphi' + 0.02 \cos \varphi'$ and a 10^{-3} relative-luminosity offset switched on. The estimator is the spin-state ratio fit of §3 with the $\sin 2\varphi'$ term; Δ follows with the MC bin-centering factor $K = \Delta_{\text{model}}(x_c, Q^2_c)/A_{\text{model, reco bin}}$, migration included.

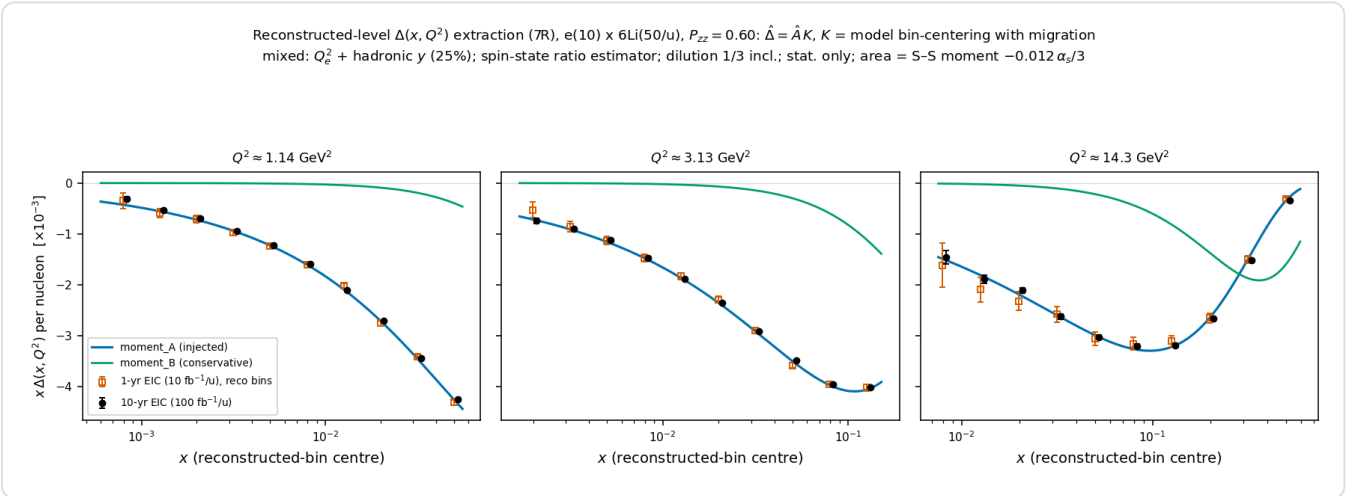
SWEET SPOT (RECO BIN)	N, 1 YR (BOTH FILLS)	PURITY	EFFICIENCY	$D = A_{\text{RECO}} / A_{\text{TRUE BIN}}$	$\hat{A} \pm \Delta \hat{A}$ (1 YR) [10^{-3}]	RECO-BIN TRUTH	$\Delta \hat{A}$ (10 YR)	ΔA OF MONEY PLOT 5
1: $x = 0.056$, $Q^2 = 1.14$	1.7×10^8	0.65	0.42	0.907	10.50 ± 0.12	10.34	0.038	0.18
2: $x = 0.022$, $Q^2 = 1.14$	2.8×10^8	0.64	0.60	0.986	6.51 ± 0.10	6.60	0.030	0.15
3: $x = 0.141$, $Q^2 = 3.13$	7.6×10^7	0.66	0.37	0.910	11.58 ± 0.18	11.71	0.058	0.26
4: $x = 0.141$, $Q^2 = 14.3$	2.8×10^7	0.68	0.64	0.969	9.52 ± 0.30	9.32	0.095	0.45

Table 2 — money plot 5R: the four sweet-spot super-bins in reconstructed (x , Q^2), mixed method with the documented 25% hadronic- y resolution [18,19]. Efficiency = fraction of the true-bin rate reconstructed in the bin after ϵ_{eID} ; D = the rate- and ϵ_{eID} -weighted, φ' -diluted model amplitude of the events in the reconstructed bin over the rate-weighted amplitude of the true bin. The last column is the single-fill, truth-binned error of the projection report.

Three observations. (i) The fit is unbiased at the reconstructed level: every $\hat{A}$ agrees with its reco-bin truth within the statistical error while the acceptance harmonic and the luminosity offset are on, and the $\sin 2\varphi'$ coefficients are consistent with zero. (ii) The two edge bins (spots 1 and 3 at $y \approx 0.01$) lose about 60% of their events to the reconstructed $y \geq 0.01$ cut and their amplitude reference shifts by 9%, the accepted half sitting at lower true x ; K knows both, and the model's x -dependence is still resolved bin by bin (money plot 7R, best bins $\delta \Delta = 1.0 \times 10^{-3}$ at $Q^2 = 1.14$ and 0.5×10^{-3} at 3.13 GeV^2 in year 1, purities ≈ 0.55). (iii) The errors are *smaller* than the truth-binned single-fill projection, 0.64–0.70 of the money-plot-5 values, because the $1.5\times$ gain of the $m = 0$ -rich fill outweighs the reconstruction efficiency of 0.37–0.64 — the number to carry into plans/07 §7.1, and it presumes the source delivers $m = 0$ -rich bunches at $|P_{zz}| = 1.2$, the same purity as the ± 1 fill at 0.6.



Money plot 5R — inclusive gluonometry at the reconstructed level ($e(10) \times 6\text{Li}50$, $P_{zz} = 0.6$, moment_A, mixed method with 25% hadronic y, EMCal energy, track angles, $\epsilon_{\text{eID}}(\eta)$, spin-state ratio of the $+0.6/-1.2$ fills with the acceptance harmonic and luminosity offset on). Left: the acceptance-free modulation $\pi(\phi') - \hat{\kappa}$ per unit P_{zz} from the ratio, with the reco-bin truth (blue) and the fit (vermillion); annotations give purity, D and the 1-/10-year errors. Right: the amplitude in reconstructed x bins along the $Q^2 = 1.14$ GeV² slice against the reco-bin truth (blue ticks) and the model curves.



Money plot 7R — $\Delta(x, Q^2)$ from reconstructed bins. $x\Delta$ points at the three sweet-spot Q^2 slices from the same pseudo-measurements, converted with the MC bin-centering factor K (model-derived, as in money plot 7, so the points sit on the injected curve by construction and carry the statistical errors of the reconstructed analysis); 1-year (open) and 10-year (filled) draws are independent.

The bin-centering factor, fitted rather than assumed. The $\cos 2\phi'$ amplitude is *exactly* linear in Δ , so `RecoResponse.fold` is the response as an exact linear operator and `fold_shape_fit` fits a $\Delta(x)$ shape through it per Q^2 slice on a bounded grid (--unfold folded; --unfold model stays the default and reproduces every published 7R number bit for bit). Correcting with a deliberately wrong prior, the residual bias over the plotted bins falls from 22% / 24% / 99% to at most 3.5% / 7.1% / 6.9% in the three slices, and the model dependence at the four sweet spots from $(-5.0, +8.5, -9.3, +6.3)\%$ to $(-0.9, -1.2, -0.2, +0.3)\%$. The 7R bar then carries four terms: the statistics, the shape fit, the response MC (0.10–1.00% per plotted bin, median 0.29%; 0.32 / 0.27 / 0.21 / 0.11% at the four spots) and — the dominant one — the spread over the priors the tilt family cannot absorb, measured to cover the residual bias at every plotted bin of every slice.

Cross-check with the hadronic final state. Rerunning 5R with the hadronic y from the **PYTHIA 8** final state through the hadron-side response (Σ method, 50 MeV noise; `money_cos2phi_reco.py --y-source hfs --hfs-sample ...`) gives

purities 0.40 / 0.53 / 0.44 / 0.73, efficiencies 0.26 / 0.46 / 0.26 / 0.72, $D = 0.79 / 0.84 / 0.86 / 0.96$ and $\delta\hat{A} = 1.2 / 1.0 / 1.8 / 2.9 \times 10^{-4}$, the fit unbiased and the median captured Σ fraction 0.92 for the selected events. **The errors are the Table 2 errors; what moves is the purity and the dilution** — the real final state puts 0.11 of Σ into neutral hadrons, and those are the HCal's, whose resolution is four to ten times the tracker's. Table 2 is therefore the optimistic end: the amplitude sits 4–21% below the true-bin value rather than 1–9%, so more of the answer passes through the bin-centering factor. It stands as long as the calorimeter noise floor is ≈ 50 MeV, degrades in proportion to it, and is 2–3 $\times$ better at the low-energy configuration (§3).

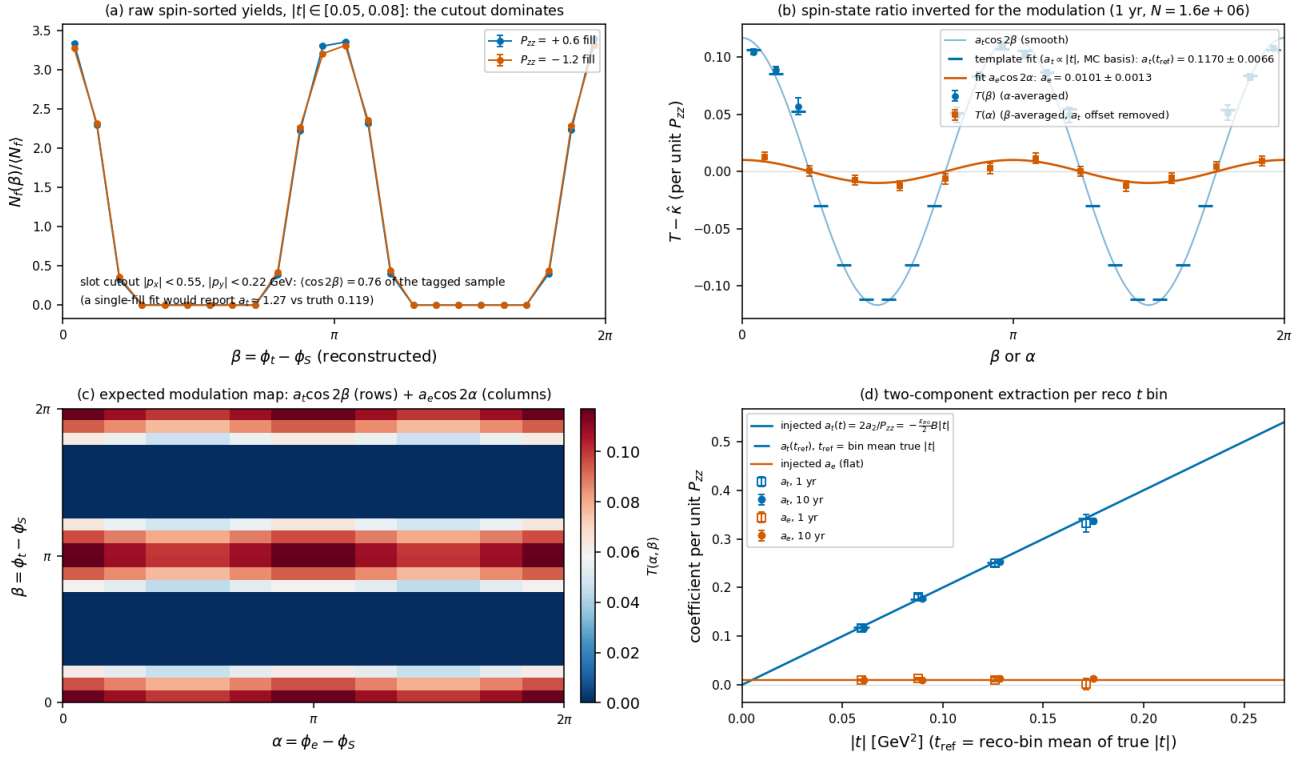
Coherent channel (money plot 6R). The coherent response samples recoils from the exponential spectrum with $x_p \in [10^{-3}, 10^{-2}]$, builds the exact four-vector and emulates the pots of §4: angular envelope $10\sigma_\theta \cdot 6p_u$ (0.22 GeV at 50 GeV/u), slot-like cutout $|p_x| < 0.55$, $|p_y| < 0.22$ GeV (width illustrative), divergence smearing. The expected counts per fill are exact α -integrals per event of $1 + u_1 \cos(\alpha - \beta) + u_2 \cos 2(\alpha - \beta) + P_t[a_e \cos 2\alpha + a_t(t) \cos 2\beta + a_m \cos(\alpha + \beta)]$, with the *true* β and t inside and the reconstructed β and $|t| = p_T^2$ for the binning. Three lessons. The anchor's convention: the deuteron paper's text fixes $d^2\sigma/d\Phi d|t| \propto 1 + 2\Sigma_n a_n e^{in\Phi}$ (its Eq. 9), so the $\cos 2\beta$ coefficient is $2a_2$, twice what money plot 6 injects ($c_2 = -(P_{zz}/2)\epsilon_{B0}B|t| = 0.072$ at $(|t|) = 0.06$ GeV², $P_{zz} = 0.6$; `coherent.cos2phi_coefficient_deformation`). The basis: the cutout leaves the in-bin β distribution far from uniform and correlates β with $|t|$ within a reco t bin, the blind directions holding the larger- $|t|$ recoils, so the fit uses the acceptance-weighted basis from the response — $\langle \cos 2\beta \rangle$, $\langle \cos \beta \rangle$, $\langle \sin \beta \rangle$ per bin plus the template $\langle (t/t_{\text{ref}}) \cos 2\beta \rangle$ of the assumed linear shape, plans/06's two-component fit $a_2(t) = c_{\text{def}}|t| + a_g$ in MC-response form; the analytic uniform-bin basis biases the deformation coefficient by 3–6% in the lowest t bin (1.25-aspect cutout), the template closes it. The geometry: the slot concentrates the recoils near $\beta \approx 0$ and π , where $\cos 2\beta$ hardly varies, so the deformation harmonic's statistical error is three times a square cutout's at similar acceptance (0.007 against 0.002 in the first $|t|$ bin at one year) while a_e is untouched — the pot geometry is a design lever for the deformation measurement and irrelevant for the exotic-glue one.

The same chain on the measured aperture. Table 3 and money plot 6R are the angular envelope with the slot cutout; the measured aperture (§4) is a second, tighter constraint, and a track must clear both (`reco.RP_APERTURE_MEASURED` , `money_cos2phi_coherent_reco.py --rp-aperture measured`). Carried through at the *low-energy* configuration, the acceptance falls 37.7% $\rightarrow$ 1.42%, $N_{\text{tag}} 8.3 \times 10^6 \rightarrow 3.1 \times 10^5$ and the acceptance-induced $\langle \cos 2\beta \rangle$ goes **+0.426 $\rightarrow$ -0.772**; the surviving window is $|t| = 0.061\text{--}0.273$ GeV², two $|t|$ bins instead of four (the higher ones no longer separate the seven harmonic columns), and δa_t degrades 0.0014 $\rightarrow$ 0.0482 and 0.0026 $\rightarrow$ 0.0150, a factor 6–34. a_e survives: 0.0073 ± 0.0045 and 0.0091 ± 0.0045 against an injected 0.0100. At the mid and top configurations the aperture leaves no accepted recoil in the binned window at all. The deformation measurement is therefore a low-energy one with a coarser t binning than Table 3, and a_e the robust half of the channel.

RECO $ t $ BIN [GeV ²]	N_{TAG} , 1 YR	$A_T(T_{\text{REF}})$ INJECTED	FIT, 1 YR	FIT, 10 YR	A_E FIT, 1 YR (INJ. 0.010)	A_E FIT, 10 YR
0.05–0.08	1.6×10^6	0.119	0.117 ± 0.007	0.1161 ± 0.0021	0.0101 ± 0.0013	0.0100 ± 0.0004
0.08–0.12	7.4×10^5	0.176	0.181 ± 0.005	0.1772 ± 0.0014	0.0130 ± 0.0018	0.0095 ± 0.0006
0.12–0.17	1.3×10^5	0.252	0.250 ± 0.008	0.2533 ± 0.0024	0.0106 ± 0.0043	0.0137 ± 0.0014
0.17–0.25	1.8×10^4	0.343	0.331 ± 0.018	0.337 ± 0.006	0.003 ± 0.012	0.013 ± 0.004

Table 3 — money plot 6R: template fit of the two-azimuth spin-state ratio per reconstructed $|t|$ bin (coefficients per unit P_{zz} ; $\epsilon_{B0} = -0.08$, $B = 50$ GeV⁻²; $u_1 = 0.05$, $u_2 = 0.02$ at the ZEUS 1σ bounds [13]; $N_{\text{tag}} = 2.7 \times 10^6$ in year 1 for the angular envelope and the slot-like cutout, against 1.1×10^7 for the constant 0.20 GeV cut). The cutout's fake $\langle \cos 2\beta \rangle$ is +0.77 — a single-fill fit would report $a_t \approx +1.3$ for a true 0.12 — and the ratio removes it entirely. The fit carries the three parity-forbidden sin harmonics as a null test (§4): all twelve coefficients are consistent with zero, and their errors resolve a Roman-Pot azimuthal roll to 5–8 mrad per $|t|$ bin in year one.

Coherent $e^6\text{Li} \rightarrow e^7\text{Li}(\text{g.s.})$ at the reconstructed level (6R), $e(10) \times 6\text{Li}(50/\text{u})$, $P_{zz} = 0.60$: $\sigma_\theta = 73 \mu\text{rad}$ ($10\sigma_\theta A p_u = 0.22 \text{ GeV}$), slot-like rectangle cutout $|p_x| < 0.55$, $|p_y| < 0.22 \text{ GeV}$
 $N_{\text{tag}} = 2.7\text{e}+06$ (1 yr) vs $1.1\text{e}+07$ with the constant 0.20 GeV cut; deformation $c_2 = 2a_2$ (Eq. 9 convention), $a_e = 0.010$
 $u_1 = 0.05$, $u_2 = 0.02$ (ZEUS LPS 1σ bounds); two-fill ratio fit with the MC template basis; statistical errors only



Money plot 6R — the coherent channel at the reconstructed level. (a) Raw spin-sorted yields versus the reconstructed recoil azimuth β in the lowest $|t|$ bin: the cutout shape dominates both fills identically (the recoils near $\beta \approx 0$ and π clear the slot). (b) The modulations $T(\beta)$ and $T(\alpha)$ from the spin-state ratio ($T(\alpha)$ with the a_t offset removed), the template-fit prediction per β bin (ticks) and the smooth $a_t \cos 2\beta$ and $a_e \cos 2\alpha$ curves. (c) The expected (noise-free) modulation map in (α, β) : rows from the deformation term, columns from the gluon-transversity term. (d) $a_t(t_{\text{ref}})$ and a_e per reco $|t|$ bin (1 yr open, 10 yr filled) against the injected curves.

7.1 • Known limits and systematics

Known limits of these numbers. The hadronic y of Table 2 is a parametrized 25% Gaussian — an absolute 50–125 MeV on $\Sigma_h = 0.2\text{--}0.5 \text{ GeV}$ at the sweet spots — so it rests on the real detector's floor at that Σ_h ; u_1 and u_2 are generated at the ZEUS 1σ bounds [13], and the slot dimensions and the ϕ' -efficiency harmonics are illustrative magnitudes; radiative corrections and backgrounds are absent, and the 6R sample is background-free and Z-identified by assumption; the 7R points are centered with the injected model unless the folded fit is used; and the template assumes the linear t-shape it fits, so a different shape is a model systematic to be varied. The fitted coefficients refer to the unsmeared truth: the α -integrals are analytic per event and the β smearing is absorbed in the MC basis. The sign of the deformation term is positive for $P_{zz} > 0$ and $\varepsilon_{B0} < 0$, the predicted flip relative to the deuteron, with the Eq. (9) convention read from the paper's own text.

Detector nuisances, measured. Each is switched on in the response and the amplitude converted with the nominal K, so the quoted number is the shift of Δ (money_cos2phi_reco.py --syst-scan, common random numbers; the response's own MC noise floor is 0.13–0.21% at the four spots). A 1% electron energy-scale error moves Δ by 0.2–1.4%, the larger of the two energy levers since $d \ln x / d \ln E' = 2 - y$ for the Σ method against $-(1 - y)$ for the hadronic scale — the Gaussian y stand-in never sees E' and gives exactly 1, understating this systematic by about two. Generating at a 30% hadronic resolution and correcting with the 25% response shifts Δ by 0.5–1.3%. An η slope of 0.05 per unit on ε_{eID} gives 0.02%: the estimator is a weighted mean of per-event amplitudes, so a smooth efficiency shape is almost exactly null and a flat one exactly null. Replacing the backward-endcap calorimeter specification by the Yellow Report η table changes nothing at spots 1–3, whose electrons are backward, and 0.5% at spot 4, whose $\eta = -1.64$ reaches the transition region. All four sit

below the 3–11% model dependence of an assumed K (moment_A/moment_B/toy at the four spots), itself larger than the 10-year statistical errors unless K is fitted through the response (above).

The systematic the ratio cannot cancel (docs/code_review_2026-08-25.md §5.1). The ratio cancels only a *common* acceptance, and the ϕ' efficiency of Table 2 is common to both fills. A difference $\delta\epsilon_2$ of the $\cos 2\phi'$ harmonic between the $m = \pm 1$ -rich and $m = 0$ -rich samples enters $\hat{A}$ as $\delta\epsilon_2/(P_+ - P_0)$ (`reco.fill_acceptance_bias` , verified against the pseudo-experiments): $\delta\epsilon_2 = 10^{-3}$ fakes 5.6×10^{-4} , 5% of the sweet-spot amplitudes but 4.6 one-year statistical errors and 14 ten-year ones. The coherent counterpart is *tighter*: a naive $\delta\langle \cos 2\beta \rangle / (P_+ - P_0)$ gives 1.3% of a_t for a 1% fill-to-fill change of the vertical envelope, but the fit gives +19% for a 10^{-3} change and +169% for 1% — under the slot only half the β bins are live and the t-shape template is 99% anti-correlated with the constant there, so a shape perturbation is amplified about a hundredfold. Bunch-by-bunch alternation, or $\approx 10^{-4}$ envelope stability, is therefore a requirement of the measurement; failing that, the higher- $|t|$ bins are far less exposed (+3.9% and +0.04% at the same 10^{-3} for $|t| = 0.08$ –0.12 and 0.12–0.17 GeV²). The EMCal term 2%/√E $\oplus$ 1% is the backward specification applied at all η (the sweet-spot electrons are backward, the high-x bins of the amplitude panels are not). u_1 , u_2 and the luminosity shares are exactly known in Table 3, but as fit arguments (`measure_coherent(u_coeffs_assumed=, lumi_assumed=, responses=)` , `money_cos2phi_coherent_reco.py --u2-assumed --rel-lumi-offset --envelope-split`), which is where the sensitivities above come from. The relative luminosity is second order in both channels: inclusively a 2% offset moves the amplitude by 0.7% and a 10^{-4} offset by 3×10^{-5} , and 10^{-3} moves a_t by 0.08%.

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[20] M. Arratia, D. Britzger, O. Long, B. Nachman, *Reconstructing the kinematics of deep inelastic scattering with deep learning*, Nucl. Instrum. Meth. A 1025 (2022) 166164, arXiv:2110.05505 (Table 1: all classical methods; Fig. 5: RMS of Q², y, x versus y for the ATHENA fast simulation; Sec. 5: acceptance and noise effects on the hadronic methods at low y). Related: R. Aggarwal, A. Caldwell, *Kinematic fitting of neutral current events in deep inelastic ep collisions*, JINST 17 (2022) P09035, arXiv:2206.04897. Local copies of [9], [12]–[20] are indexed in `refs/README.md` and `refs/refs_dict.json`.

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Appendix A · Revision history

DATE	REVISION
2026-08-24	First version: what is measured, the covariant azimuth and the head-on frame, the inclusive and coherent chains, the audit of the simulation and the recommendations (Figures 1–2, Schematics 1–3, <code>polligen.reco</code>). Same day: §7, the reconstructed-level pseudo-experiments of WP3 (money plots 5R, 7R, 6R from <code>polligen.recopseudo</code> , the acceptance-weighted template basis, the anchor's 2a _z convention).
2026-08-25	Revised against the reference copies in refs/: the ePIC Roman-Pot slot geometry and beam-effect resolution [17], the ZEUS bounds on the diffractive harmonics [13,14], the exact spin-1 harmonics at finite γ [12], the anchor's frame and convention [9]; the hadronic-y resolution set to the documented 25% [18–20] and 5R/7R rerun; the limits of §7.1 from the code review. Then the hadronic final state itself (WP3-HFS): <code>polligen.hfs</code> , the hadron-side response and the (x, Q ²) event library; §3, Figure 3 and the §7 cross-check.
2026-08-26	Three open items become measurements and one input moves a result. The hadronic final state is PYTHIA 8 , 12 M events over the three beam configurations: §3, Figure 3 and finding (1) re-derived on it — δy/y = 0.32 / 0.22 / 0.28 / 0.11 at the sweet spots against the toy stand-in's 0.28 / 0.17 / 0.24 / 0.07, purity 40–73% against the 25% Gaussian's 64–68%. The Roman-Pot aperture is measured with an intact ⁶ Li through the ePIC geometry — it opens horizontally, the opposite aspect to the slot §4 assumed — and carried through the coherent chain (§4, §7). The bin-centering factor has a fitted alternative , <code>RecoResponse.fold</code> and <code>fold_shape_fit</code> (§7). R is the published SLAC/E143 R1998 fit [21]: F ₂ cancels exactly and Δ/F ₁ moves +16.6 / +18.0 / +4.7 / −4.4% (§3). The O(γ ²) b ₁ leakage bound of §2 is corrected upward by ≈ 6.9× to the full Cosyn Eq. (17e) combination (plans/08 D2). Editorial: the audit table of §5 and the work list of §6 give way to what the chain contains and what is left to build; the item-by-item status is plans/08, and each validity limit now stands beside the number it qualifies.

Conventions: head-on frame with the ion along +z and the electron along −z; θ' is the scattered-electron angle from the electron beam; per-nucleon (x, y, s) throughout; whole-nucleus four-momenta where a mass matters. Beam divergences are the proton values implied by the documented Roman-Pot cuts (73 and 149 μrad); the lithium optics are undocumented (plans/04 #11). Built by `reports/build_report.py`; figures from the `evgen/scripts` named in the text; display mathematics typeset at build time.